



Calibration: What is it and why it matters? An introduction to the special issue on calibrating calibration

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ARTICLE INFO

Article history:

Received 23 October 2012

Accepted 23 October 2012

Keywords:

Calibration

Self-regulation

Metacognition

Judgments of confidence

ABSTRACT

The burgeoning literatures on students' metacognitive judgments, self-regulation, self-efficacy beliefs, strategic processing, and conceptual change have focused new attention on the nature and measurement of calibration. Calibration, the distance between perceived and demonstrated levels of understanding, capability, competence, or preparedness, is foundational to these critical areas of inquiry. Yet, not enough is known about the nature of calibration or about the best means of gauging the distance between what students' believe and what they show in their cognitive actions and performance. It is the goal of this special issue on "Calibrating Calibration" to provide educational researchers with critical theoretical and methodological information on this pivotal topic.

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At its simplest, calibration within the domain of human learning reflects the degree to which individuals' judgments about their understanding, capability, competence, or preparedness correspond to the understanding, capability, competence, or preparedness they actually manifest (Lichtenstein, Fischhoff, & Phillips, 1982). The degree of correspondence between such a priori judgments and post hoc evidence is a mark of good to poor calibration. For instance, if a student feels that she has well understood a text passage and, in fact, demonstrates her comprehension when responding to questions, we would say that her calibration was good in that instance. However, if another student determines that he has studied sufficiently to do well on tomorrow's geometry exam, but ends up receiving a disappointing score, we would judge his calibration as poor. Poor calibration can also be signified by individuals' under-confidence as much as by the overconfidence seen in our geometry student. Thus, the student who underrates her capability to score well on a Biology exam also manifests poor calibration.

There has been evidence forwarded within the extant literature that students are not especially good at accurately judging themselves and their preparedness against some perceived standard or future performance expectation (Bol & Hacker, 2001; Klassen, 2002). Although quality of calibration judgments can be associated with general cognitive ability, academic achievement, or the level of expertise of those making such determinations, and manifest differential patterns by cultural group or academic domain (Bol, Hacker, Walck, & Nunnery, 2012; Chiu & Klassen,

2010), calibration concerns afflict all of us at some time and in some way. It is not always easy to answer the questions: Do I understand this? Am I cognitively ready for the task ahead? How much more effort is required for me to be adequately prepared? Yet, such metacognitive judgments, while rarely simple or straightforward, form the basis for calibration and are instrumental to individuals' self-regulatory judgments and strategic behaviors.

As someone who has spent decades empirically studying learning and academic performance, metacognitive, and strategic processing, and as a teacher who has sought to guide students toward more informed judgments about their capabilities, competence, and preparedness, I remained inadequately informed about calibration for some time. Answers to basic questions about this construct eluded me such as: What precisely is calibration? On what basis do students determine that they understand a given task or problem adequately or that they have achieved a level of understanding necessary to complete that task or problem well? And, what interventions or modifications can be implemented to improve students' ability to form sound judgments about their capabilities, competence, and preparedness vis-à-vis their goals and the tasks at hand. My lingering questions extended directly to questions of measurement and statistics as well, including how calibration can be reliably and validly measured and whether one mode of rating or data analysis has superiority over another.

The reasons for pursuing the potential answers to such questions are not solely driven by self-interest. There is a growing interest in the topic of calibration within the broader educational research community—and for good reasons. For one, that rising interest relates to the fact that various constructs and processes foundational to human learning and development rest on the

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ability of learners to calibrate well. Take metacognition, self-regulation, self-efficacy beliefs, strategic processing, and conceptual change as cases in point. All of these research domains are widely investigated within the educational research community, as evidenced by the specific special interest groups or SIGs dedicated to these topics at EARLI and AERA, and the extensive literatures dedicated to each topic, including the special issue on self-monitoring and self-regulation recently published in *Learning and Instruction* (De Bruin & Van Gog, 2012).

So much inquiry into the aforementioned areas only serves to highlight the comparatively little that is known about the nature of calibration and the bases for students' initial judgments of their capabilities, competence, or preparedness for specific academic tasks, despite the fact that calibration is a major component in several metacognitive and self-regulatory models of learning (e.g., Winne & Hadwin, 1998). Further, calibration is particularly important because students' ability to gauge their progress and performance accurately will impact their subsequent effort and strategic behaviors. For instance, what would be the reason for students to continue monitoring their performance or engaging in strategic processing unless they recognized some discrepancy between their present understanding and that presumably required of a given task or problem? Likewise, the likelihood of conceptual change, especially intentional conceptual change (Sinatra & Pintrich, 2003) would seem predicated on whether learners are able to assess their present understanding vis-à-vis the more scientific or expert models they are presented or the conflicting and counter evidence they encounter.

But researchers' efforts to investigate calibration or to devise instructional interventions to improve students' performance judgments have been occasionally stymied by limited methodological and statistical resources. For example, there are lingering concerns as to how best to gather students' self-ratings, and what statistical procedures can be used to calculate calibration for a given dataset more appropriately or accurately. If educational researchers are going to move forward in the study of calibration, they must be well informed and well equipped with suitable measurement and statistical tools and techniques.

To address this array of theoretical, methodological, and statistical questions, I set out to put together this special issue on "Calibrating Calibration" and turned to scholars from across the globe to share their research and critical insights on calibration. The result is a diverse set of cutting edge studies that explore calibration through varied conceptual, contextual, and statistical lenses. Each of the empirical studies in this issue positions calibration within a theoretical framing (e.g., metacognition or self-regulation) in order to better contextualize the nature of students' judgments about their capabilities, competence, or preparedness and their judgment accuracy. Further, while the particular goals or purposes espoused by authors of the five empirical studies differed, along with the content domains or age of participants selected for study, they collectively sought to address the two questions at the heart of this special issue:

1. What is the nature of or bases for the judgments students make about their capabilities, competence, or preparedness?
2. How can researchers realistically and effectively measure such judgments and calculate calibration from the resulting data?

For example, in the study by Dinsmore and Parkinson (2013), undergraduate students in a research methods course read and recall two complex texts and rate their confidence in their responses using one of two approaches (i.e., 100 mm line or magnitude scale). The authors also delve into the criteria that students use to arrive at their confidence ratings. What Dinsmore

and Parkinson determine is that students rely on multiple criteria (e.g., prior knowledge, text characteristics, and guessing) when forming their confidence judgments. Further, the authors reach the conclusion that the two approaches for acquiring these judgments did not significantly affect the outcome. The authors thus weigh both the instructional and the statistical implications of their findings.

Van Loon, De Bruin, Van Gog, and Van Merriënboer (2013) also consider the relation between prior knowledge and calibration accuracy, but among primary-grade students. What the authors want to ascertain is whether the activation of inaccurate prior knowledge alters the judgments that these younger students make about their learning and performance. In fact, what we learn is that activation of inaccurate prior knowledge not only negatively affects students' learning but also manifests in poor calibrations—an outcome that has implications for research in conceptual change as well as self-regulation.

In a methodologically creative investigation, Higham (2013) examines the effect of what he calls the "plurality option" on the confidence that students' report for their responses on a multiple-choice test in psychology. The plurality option refers to students' confidence rating for their favorite response alternative on the multiple-choice item (i.e., small answer) and, simultaneously, for the remaining alternatives (i.e., large answer), along with their freedom to select which of these two ratings should serve as the basis for grading. Higham reports that students' response accuracy was higher, but their confidence lower, when they selected large answers as compared to small answers for grading. On the basis of these findings, Higham argues for the use of the plurality option to help undergraduate students regulate their response accuracy on multiple-choice tests.

As with Higham, Hadwin and Webster (2013) investigate undergraduate students' judgments of confidence with a contextual twist. What is especially important about this investigation is that the students in the authors' study were involved in an online program meant to improve their academic performance (i.e., Personal Planning Tools). Thus, calibration was a natural component of this program where students set weekly goals for performance in one of their university courses and then judged their attainment of those goals. Overall, the authors identify a significant relation between calibration accuracy and academic performance. Moreover, using multilevel linear modeling, they report that participants' judgments of confidence are better when rating current rather than past goal attainment, although whatever improvements the authors expected from the Planning Tools program over time did not emerge.

Finally, Schraw, Kuch, and Gutierrez (2013) bring their expertise to the study of calibration by examining the dimensionality of 10 different calibration measures found in the literature. Using a common dataset and confirmatory factor analyses, the authors compare one-, two-, and five-factor solutions and show that the two-factor solution, corresponding to measures of specificity and sensitivity, is the most diagnostically efficient. Schraw et al. then offer a detailed discussion intended to shed light on the underlying processes of metacognitive monitoring that these two factors seem to reflect.

As important as these individual empirical contributions are to understanding the nature and measurement of calibration, the two concluding analyses by John Dunlosky and Keith Thiede (in this issue) and by John Hattie (in this issue) afford a more encompassing perspective on the topic. Few educational researchers are better positioned to offer critiques and insights on calibration than these international scholars—and they do just that. What they specifically share also has the potential to carry the study of calibration to the next level.

And moving the study of calibration forward to the next level is precisely what is needed. Not only for those directly invested in the investigation of this construct per se, but also for all those educational researchers who are engaged in the areas of metacognition, self-regulation, self-efficacy beliefs, strategic processing, and conceptual change; that is, a broad cross-section of the research community.

Acknowledgments

As editor for this Special Issue, I am especially grateful to Dr. Lucia Mason and her associate editors for allowing me the opportunity to bring this topic to the attention of the educational research community. I am also beholden to Dr. Mason for her ongoing guidance, input, and support in the editorial process that served to make this issue the best it could be. Along with Dr. Meghan Parkinson, who urged me to pursue this topic and offered me her insights and recommendations, I want to express my deep appreciation to the outstanding set of reviewers whose thoughtful and detailed analyses ultimately enhanced the quality of each of the contributions found in this Special Issue.

References

- Bol, L., & Hacker, D. J. (2001). A comparison of the effects of practice tests and traditional review on performance and calibration. *The Journal of Experimental Education*, 69, 133–151. <http://dx.doi.org/10.2307/20152656>.
- Bol, L., Hacker, D. J., Walck, C. C., & Nunnery, J. A. (2012). The effects of individual or group guidelines on the calibration accuracy and achievement of high school biology students. *Contemporary Educational Psychology*, 37, 280–287. <http://dx.doi.org/10.1016/j.cedpsych.2012.02.004>.
- Chiu, M. M., & Klassen, R. M. (2010). Relations of mathematics self-concept and its calibration with mathematics achievement: cultural differences among fifteen-year-olds in 34 countries. *Learning and Instruction*, 20, 2–17. <http://dx.doi.org/10.1016/j.learninstruc.2008.11.002>.
- De Bruin, A., & Van Gog, T. (2012). Improving self-monitoring and self-regulation of learning: from cognitive psychology to the classroom. *Learning and Instruction*, 22, 245–298. <http://dx.doi.org/10.1016/j.learninstruc.2012.01.003>.
- Dinsmore, D. L., & Parkinson, M. M. (2013). What are confidence judgments made of? Students' explanations for their confidence ratings and what that means for calibration. *Learning and Instruction*, 24, 4–14.
- Dunlosky, J., & Thiede, K. W. (2013). Four cornerstones of calibration research: why understanding students' judgments can improve their achievement. *Learning and Instruction*, 24, 58–61.
- Hadwin, A. F., & Webster, E. (2013). Calibration in goal setting: examining the nature of judgments of confidence. *Learning and Instruction*, 24, 37–47.
- Hattie, J. (2013). Calibration and confidence: where to next? *Learning and Instruction*, 24, 62–66.
- Higham, P. A. (2013). Regulating accuracy on university tests with the plurality option. *Learning and Instruction*, 24, 26–36.
- Klassen, R. M. (2002). A question of calibration: a review of the self-efficacy beliefs of students with learning disabilities. *Learning Disability Quarterly*, 25, 88–102. <http://dx.doi.org/10.2307/1511276>.
- Lichtenstein, S., Fischhoff, B., & Phillips, L. D. (1982). Calibration of probabilities: the state of the art to 1980. In D. Kahneman, P. Slovic, & A. Tversky (Eds.), *Judgment under uncertainty: Heuristics and biases* (pp. 306–334). New York: Cambridge University Press.
- Schraw, G., Kuch, F., & Gutierrez, A. P. (2013). Measure for measure: calibrating ten commonly used calibration scores. *Learning and Instruction*, 24, 48–57.
- Sinatra, G. M., & Pintrich, P. R. (2003). *Intentional conceptual change*. Mahwah, NJ: Lawrence Erlbaum Associates.
- Van Loon, M. H., De Bruin, A. B., Van Gog, T., & Van Merriënboer, J. J. (2013). Activation of inaccurate prior knowledge affects primary-school students' meta-cognitive judgments and calibration. *Learning and Instruction*, 24, 15–25.
- Winne, P. H., & Hadwin, A. F. (1998). Studying as self-regulated learning. In D. J. Hacker, J. Dunlosky, & A. C. Graesser (Eds.), *Metacognition in educational theory and practice* (pp. 277–304). Mahwah, NJ: Lawrence Erlbaum Associates.